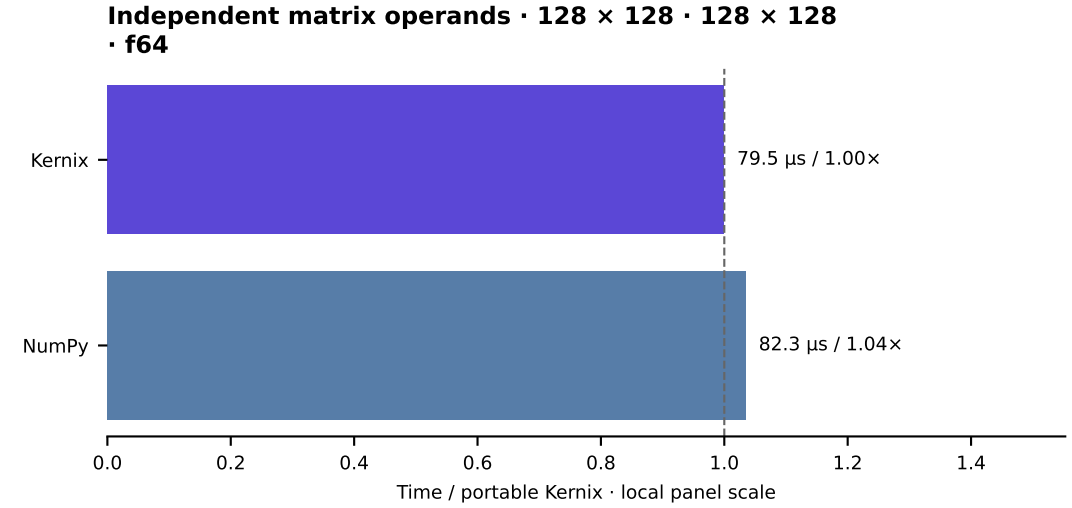
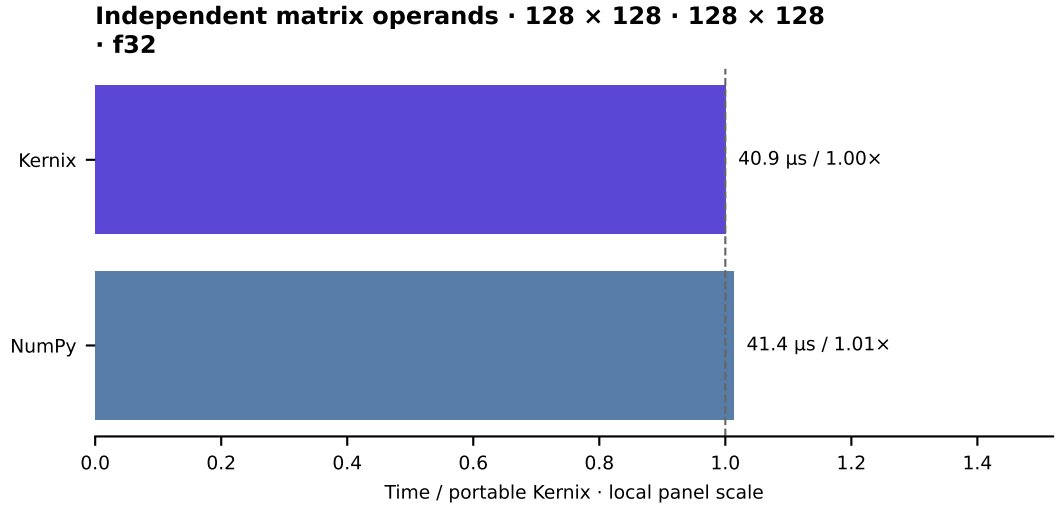
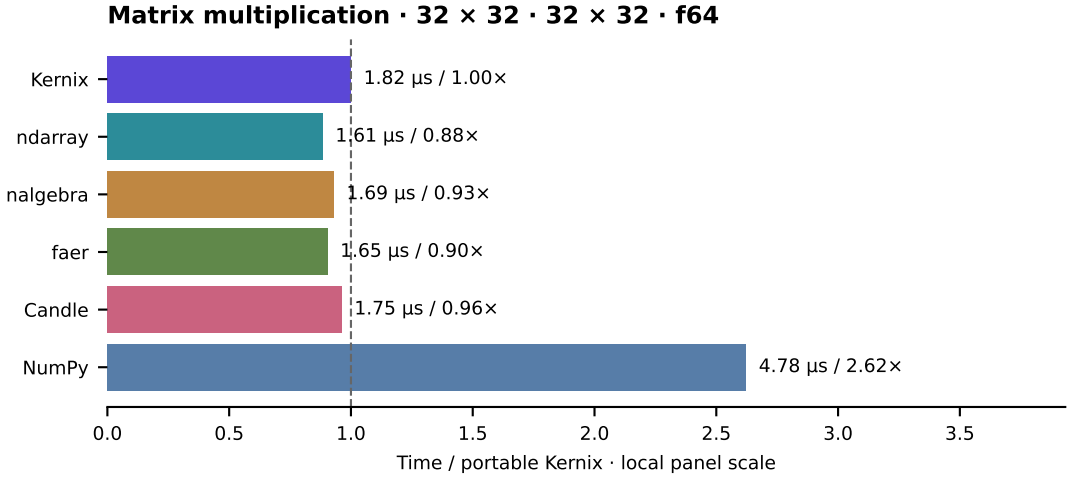
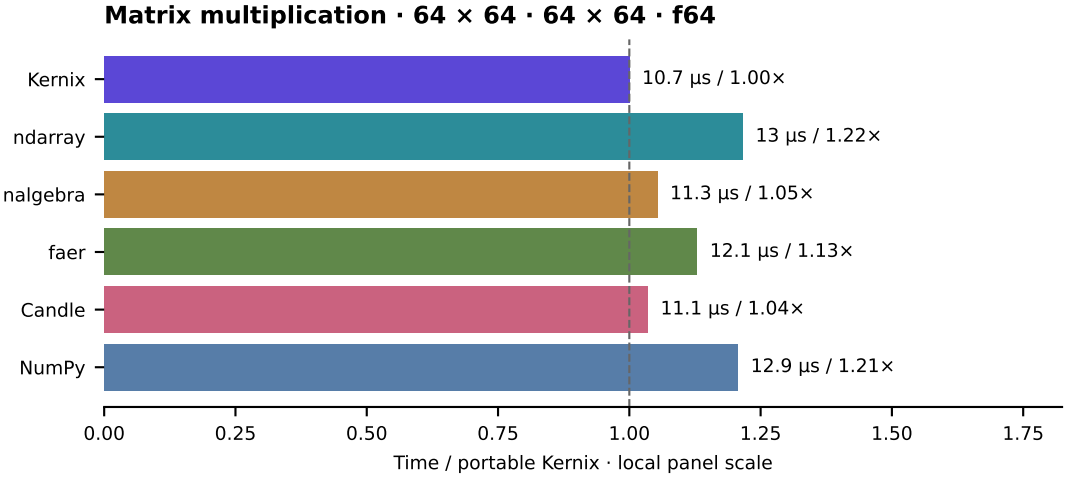
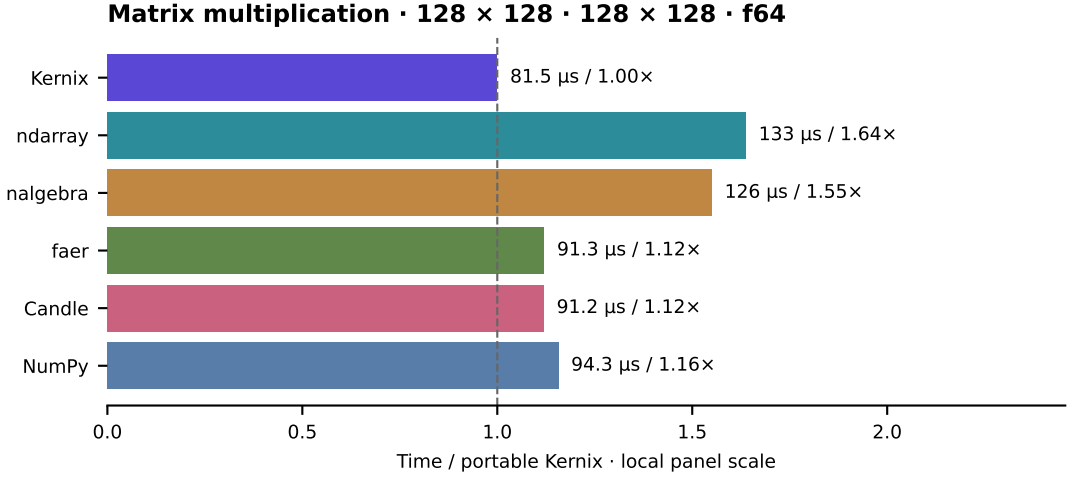
Matrix multiplication · page 1/9

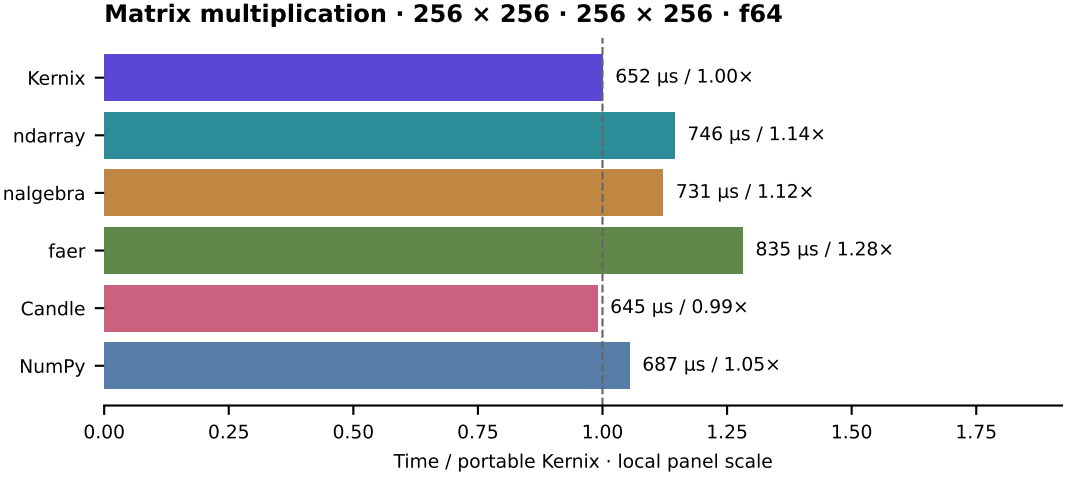




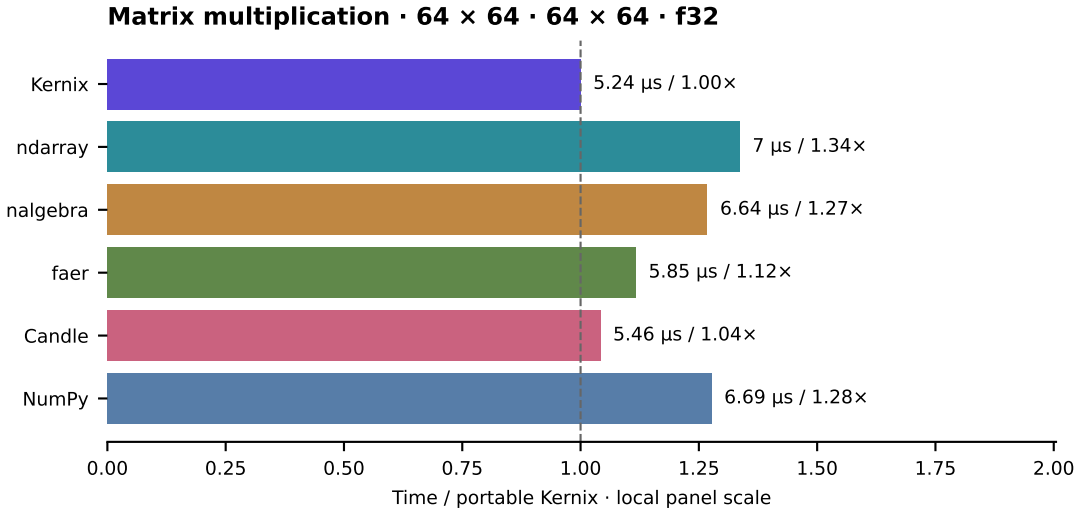


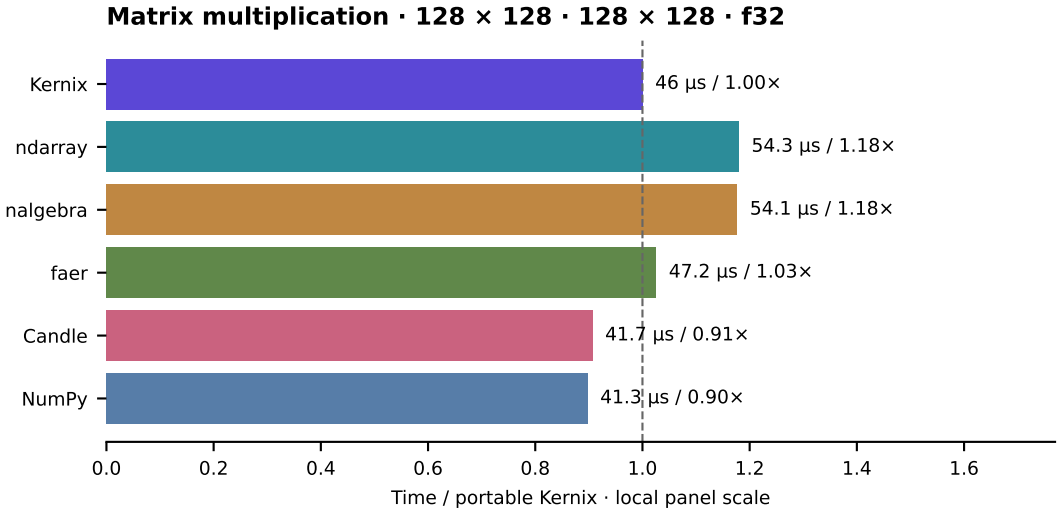


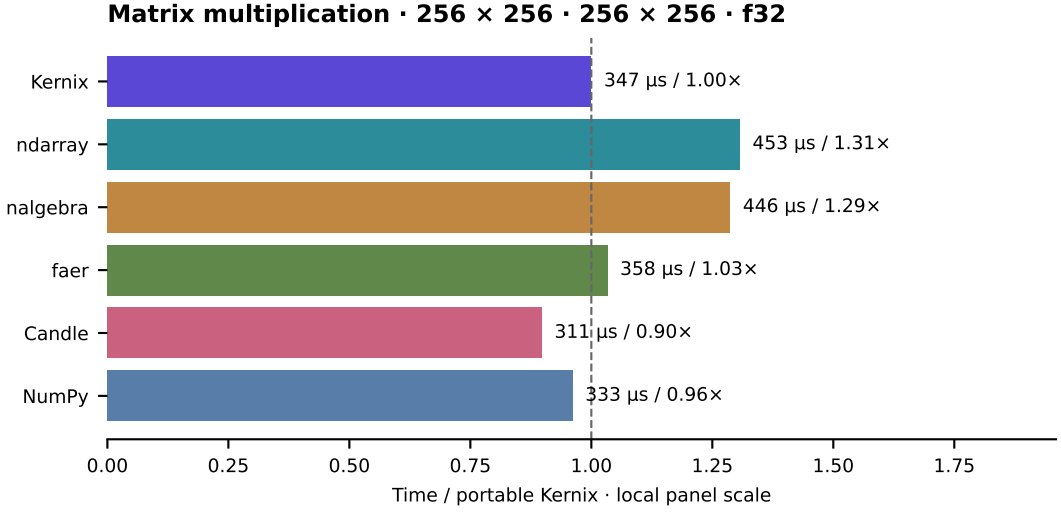


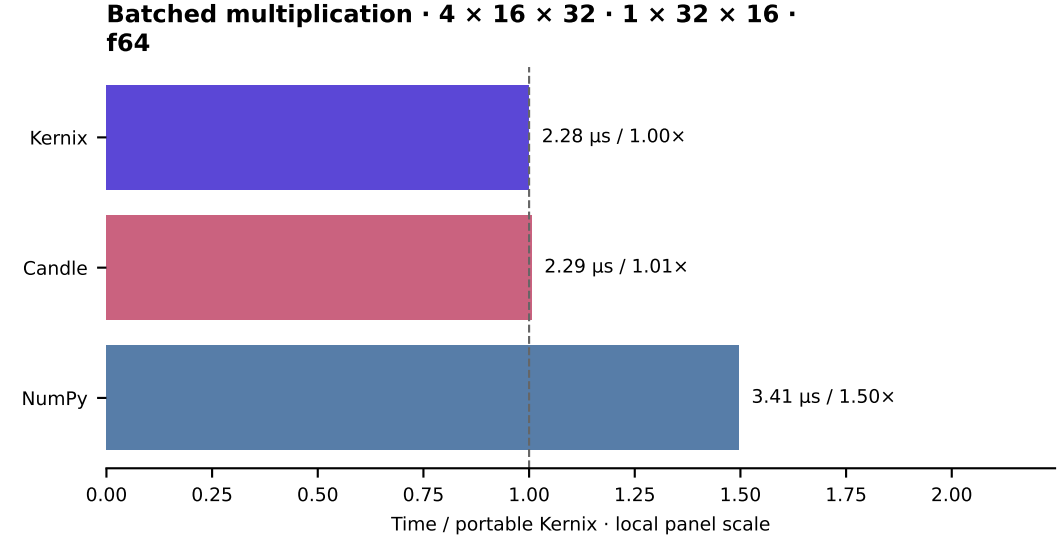


Matrix multiplication · page 2/9









Explicit parallel execution · page 3/9

Matmul · 2 workers · reused output · 64 × 64 · 64 × 64 · f64

Previous-version comparison only; no peers measured

Matmul · 2 workers · reused output · 128 × 128 · 128 × 128 · f64

Previous-version comparison only; no peers measured

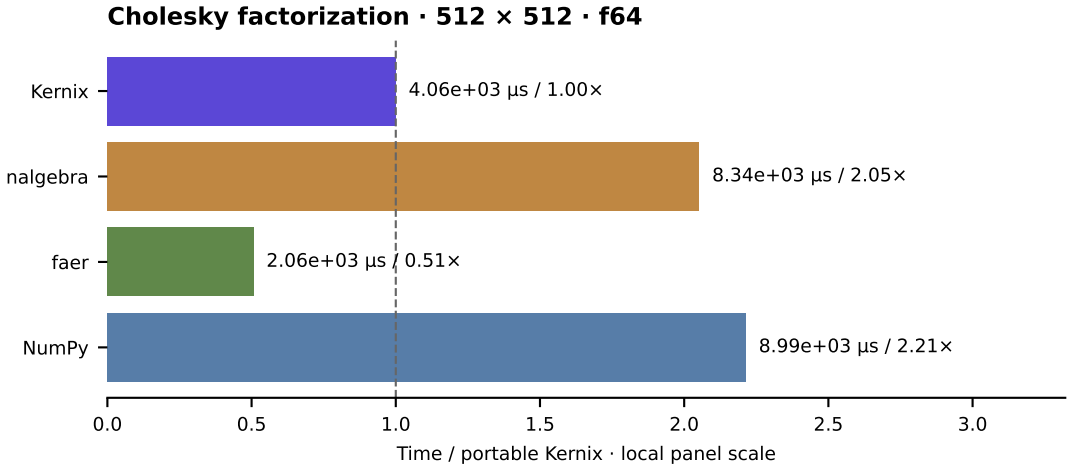
Matmul · 2 workers · reused output · 256 × 256 · 256 × 256 · f64

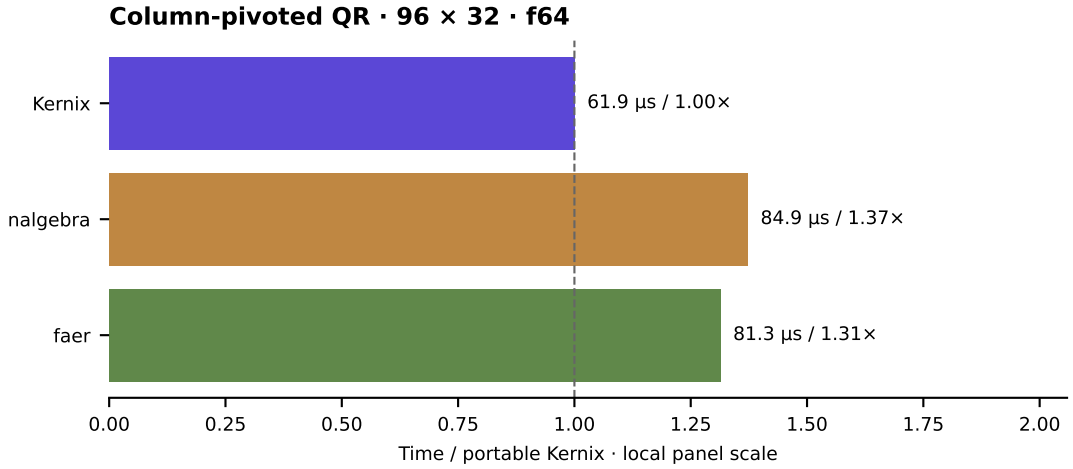
Previous-version comparison only; no peers measured

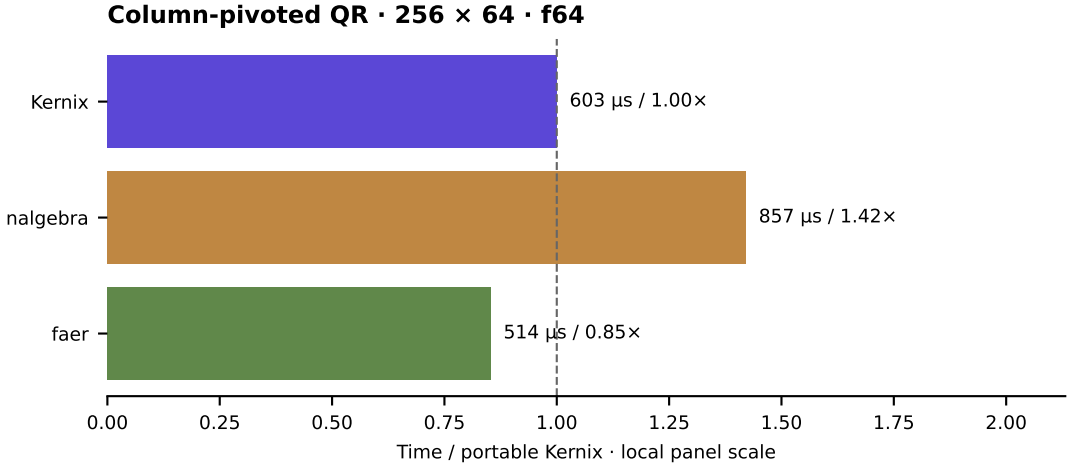
Matmul · 2 workers · reused output · 128 × 128 · 128 × 128 · f32

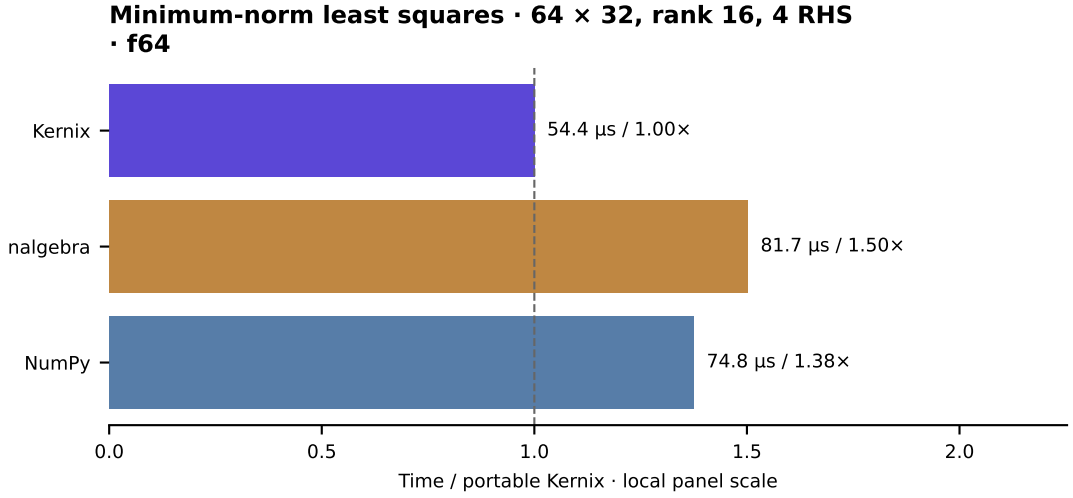
Previous-version comparison only; no peers measured

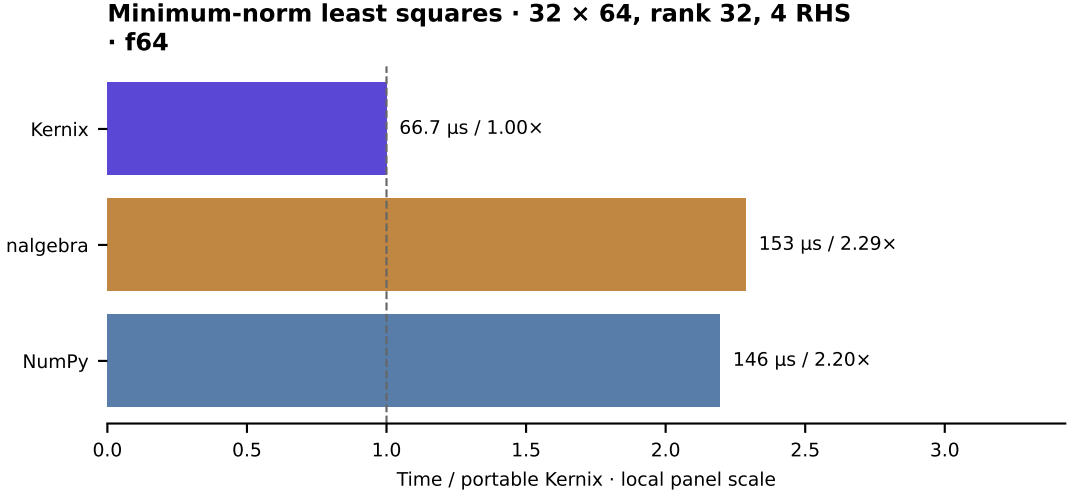
Decompositions and solvers · page 4/9

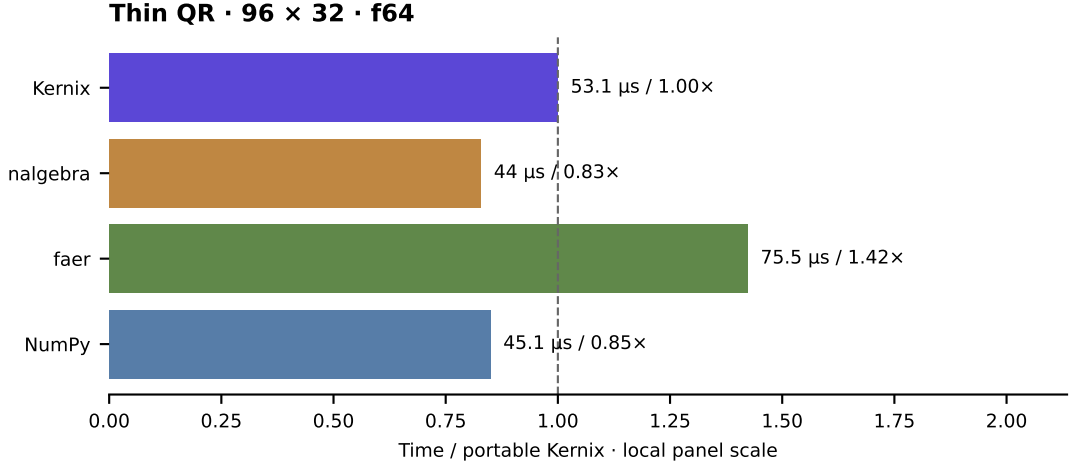




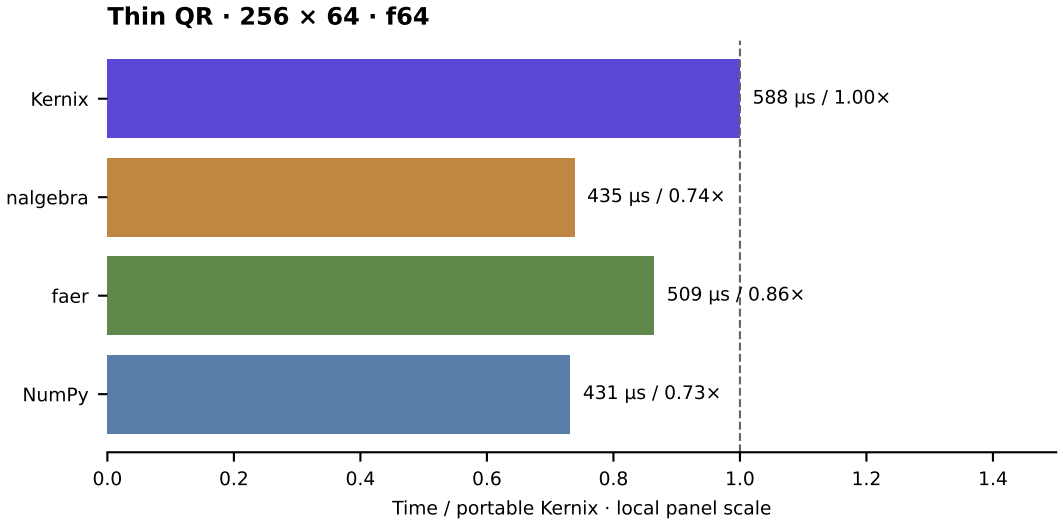


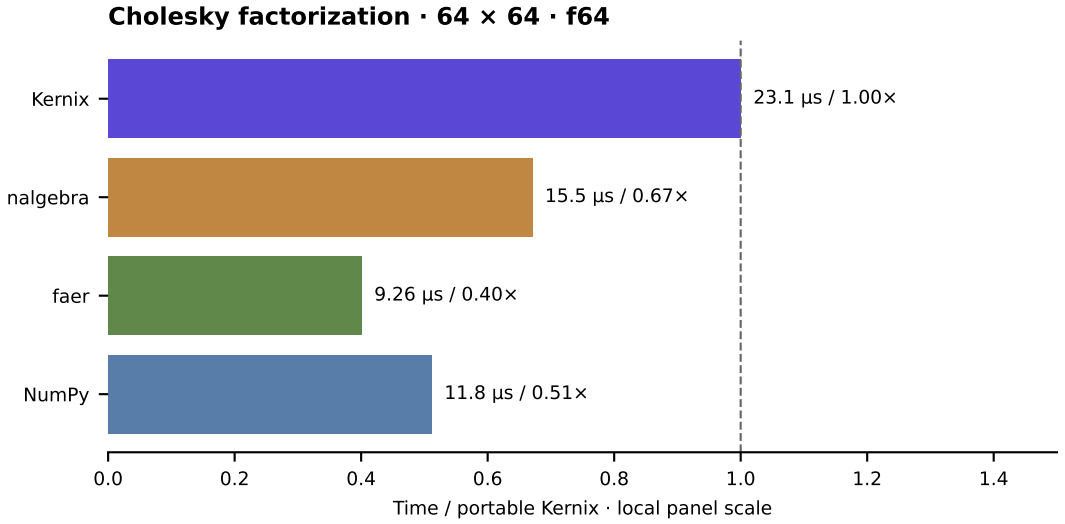


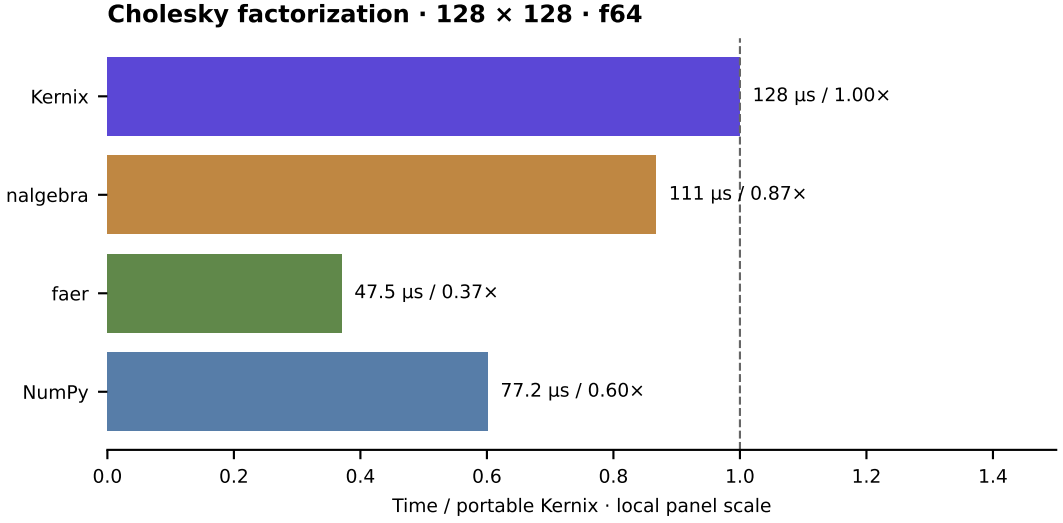


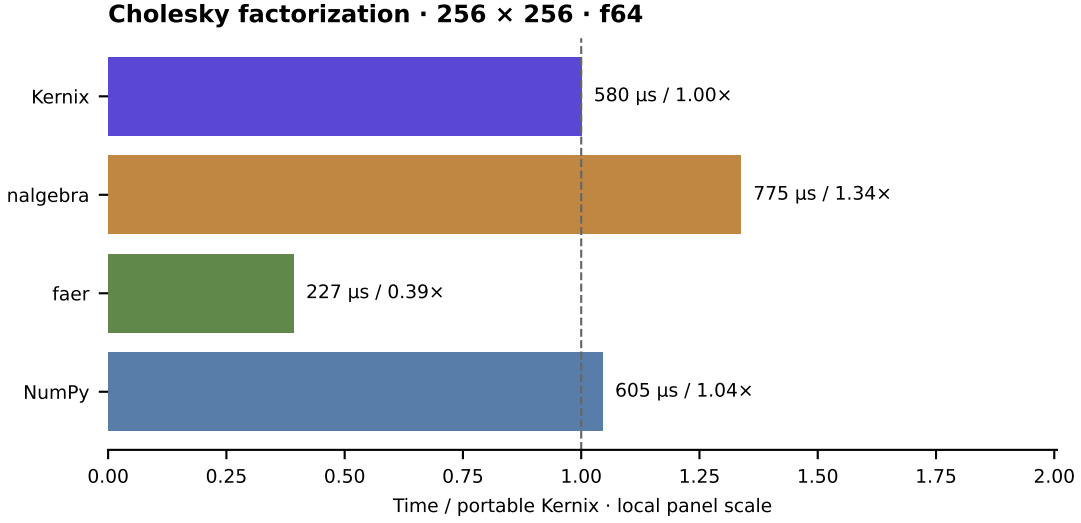


Decompositions and solvers · page 5/9



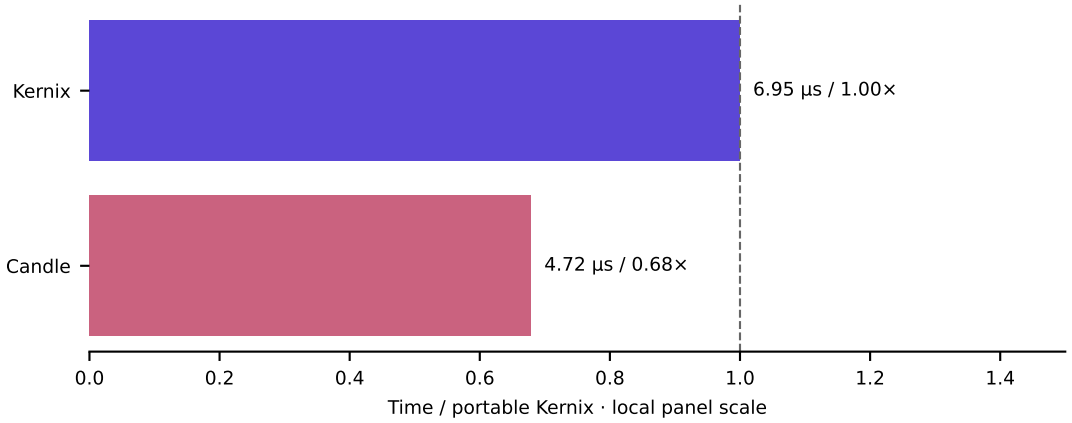




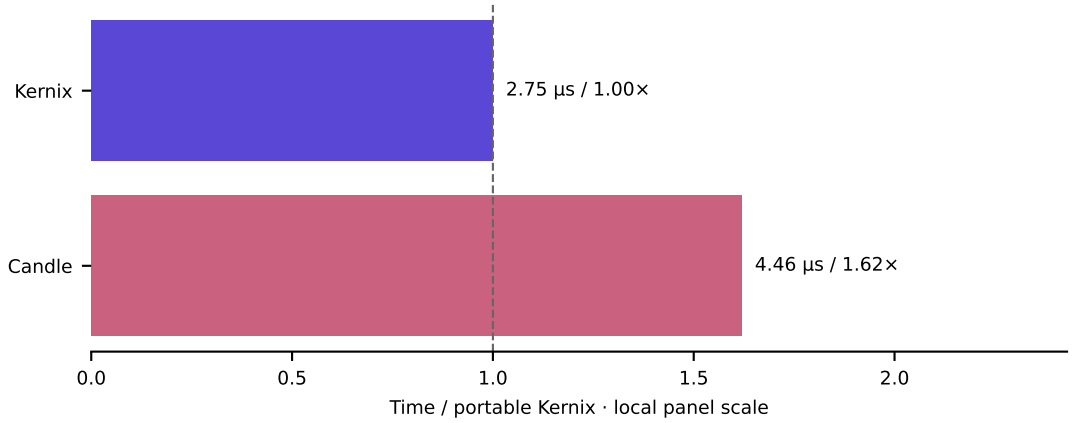


Training and autodiff · page 6/9

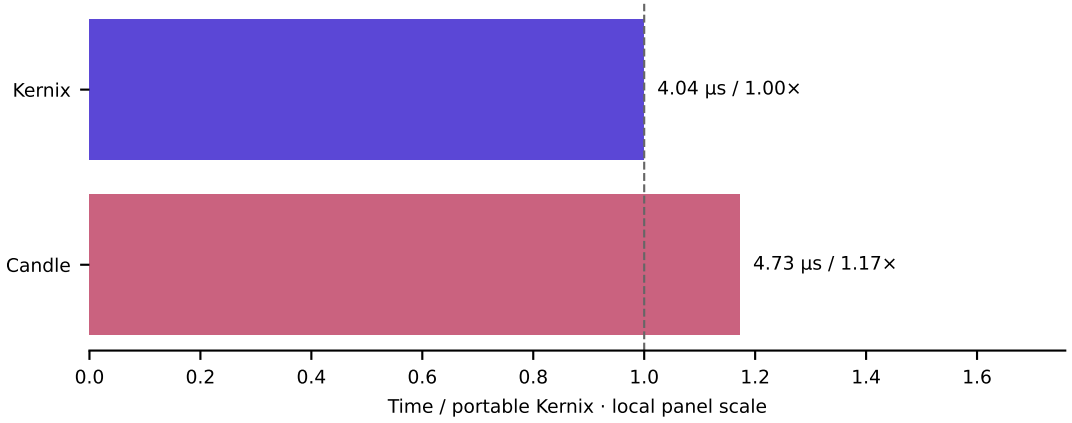
Shared-variable backward · sum(X * X), 64 × 32; prepared graph · f64



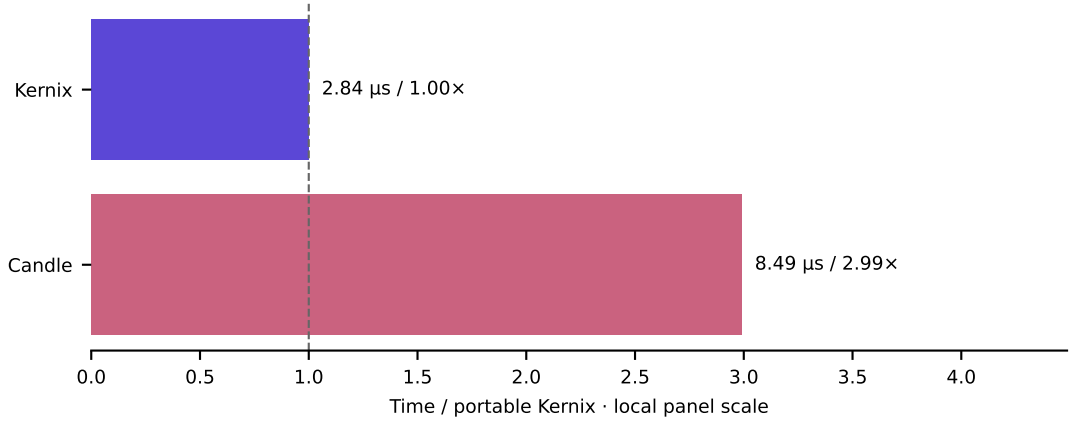
Record + backward · sum(A · W), 32 × 16 · 16 × 8 · f64



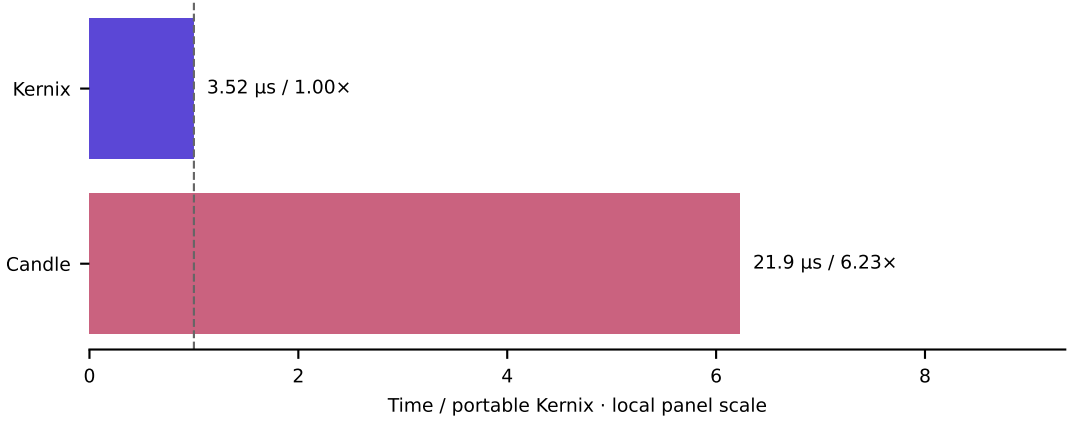
Model inference · batched normalized/sliced MLP · f64



Model recording + forward · batched normalized/sliced MLP · f64

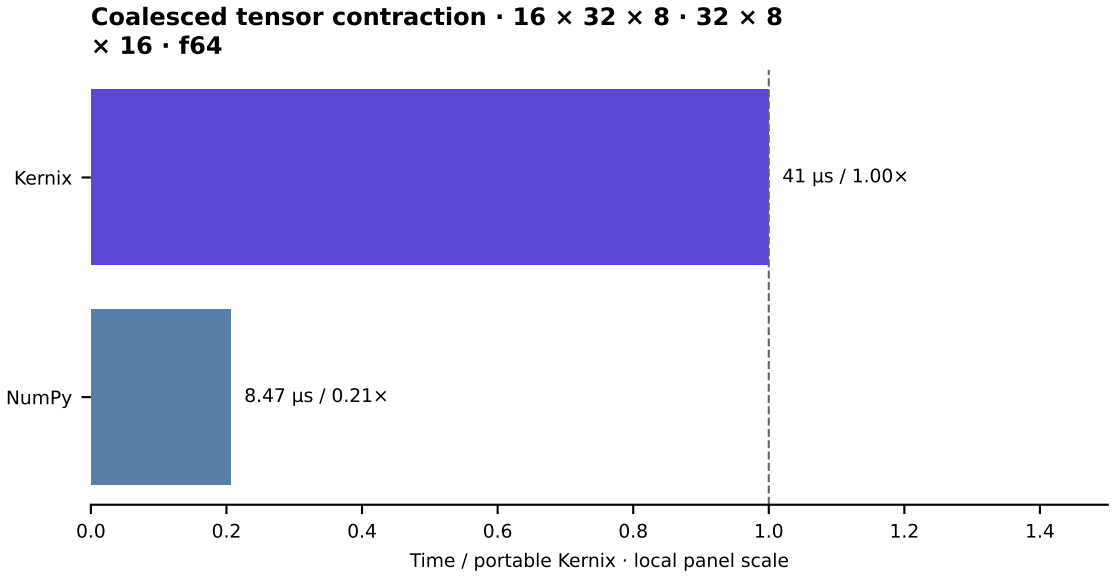


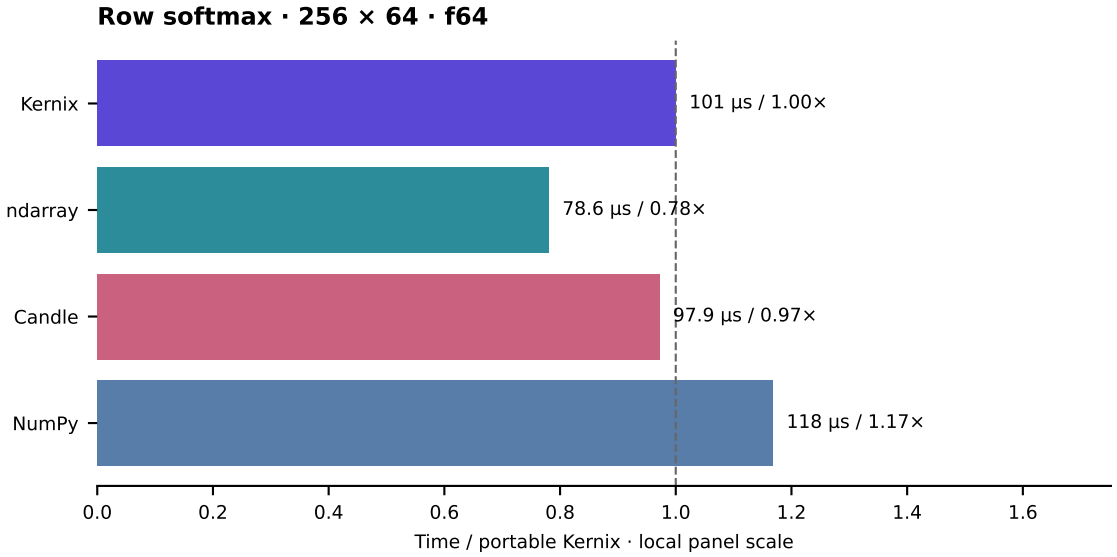
Model backward · prepared graph → fresh gradients · f64



Parameter update · transactional SGD, fixed gradients · f64

Previous-version comparison only; no peers measured


Coalesced tensor contraction · 16 × 32 × 8 · 32 × 8 × 16 · f64
Kernix
NumPy
41 µs / 1.00×
8.47 µs / 0.21×
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Time / portable Kernix · local panel scale


Row softmax · 256 × 64 · f64
Kernix
ndarray
Candle
NumPy
101 µs / 1.00×
78.6 µs / 0.78×
97.9 µs / 0.97×
118 µs / 1.17×
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
Time / portable Kernix · local panel scale

Tensor construction · page 9/9

